

Does Health Matter? A Quantitative Analysis of Cross-country Income Differences

Job Market Paper

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Abstract: Standard neoclassical growth theory requires implausibly large productivity gaps in order to explain cross-country income differences. This paper illustrates the significant role played by health factors in such accounting, and also argues that a poor health environment makes health investment less rewarding, affects the parents' attitude toward investment in children, and retards human capital accumulation and output growth. By estimating health capital across countries and calibrating the model to fit the data for the U.S. economy, I then calculate the required TFP gaps to account for the cross-country differences in relative outputs. With the introduction of the health production input, the required TFP gaps shrink substantially. The results also indicate that, while education is important in explaining cross-country income differences, the variations in the health conditions and health investment efficiencies are vital as well, especially for poor nations in sub-Saharan Africa and South Asia.

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1 Introduction

Why are some countries poor? Why are others rich? What are the forces behind the income disparities across nations? Development accounting attempts to answer these questions by focusing on the two proxies – factors of production and efficiency. Besides large income variations across countries, disparities in health performances and educational attainment are also sizable. Previous studies in development accounting focus more on the role of knowledge, such as the quantity and quality of education and the experiences of workers. This paper, instead, tackles the problem of health inequality and efficiency in producing health, and examines whether health plays a significant role in explaining income differences across countries.

Figures 1-3 plot the life expectancy at birth against the log real GDP per capita, the average years of schooling against the log real GDP per capita, and the life expectancy at birth against the log real GDP per capita for an overview of the inequality in income, education and health across countries. To further examine the worldwide inequality in income and health, I report the survival rate to age 25 and life expectancy at birth by income level for 117 countries using data circa the year 2000 in Table 1.¹ I also report the average growth rate per capita during 1970-2007 and the average relative real GDP per worker in 2000 with the normalization that the real GDP per worker of the U.S. in the year 2000 is equal to one. From Table 1, it can be observed that there is a notable difference in the performance of health and the economic growth rate between countries with an output per capita of less than and more than 10 percent of that of the U.S. For countries with an income level between five and 10 percent of that of the U.S., the average life expectancy at birth was found to be 61.9 years, and nearly 90 percent of live born babies survived to see their 25th birthday. The average economic growth rate during 1970-2007 was 1.42%, which was lower than the average growth rate for most income groups. However, countries with an income level of less than five percent of that of the U.S. had an average life expectancy of 51.5 years, and only 80 percent of their live born babies lived to see their 25th birthday, representing a 17 percent and 10 percent more attrition in human lives compared to countries with an income of 30 percent or more of that of the U.S. and 5-10 percent of that of the U.S., respectively. A more striking fact is that the average growth rate per capita of these poor countries during 1970-2007 was only 0.38 percent, being ranked the lowest among all income groups. This indicates that these countries suffered from the problem of persistent poverty.

How does health affect individuals and economic output levels? Several effects of health on individual behavior and economic activities have been identified in the literature of macroeconomics and health economics. Those effects can be summarized in three channels: (1) the direct effect of health; (2) the within-generation indirect effect of health; and (3) the intergenerational indirect

¹ A list of the countries under investigation is given in the Appendix.

effect of health. The first channel, the direct effect of health, is that better health brings higher productivity both in work and education. Healthier workers are usually more energetic and can work longer. Better health also promotes clearer thinking and faster learning. Furthermore, healthy and energetic workers tend to have a more optimistic outlook of the future and are willing to work harder. Hence, given the same labor and study time as less healthy workers and students, healthier workers can produce more output and accumulate more human capital in the form of working experience, and healthier students can receive a better education.

The second channel, the within-generation indirect effect of health, refers to all the effects indirectly resulting from health within a single generation. For instance, a longer life expectancy due to health improvement will induce more schooling, since people enjoy a longer lifespan to amortize the cost of education. These increases in both education and health lead to a higher level of human capital, which increases the marginal product of physical capital, and hence will lead to a rise in investment and accumulation of physical capital. Further, if people expect to have a long retirement period due to a longer life expectancy, they will save more.

The last effect, the intergenerational indirect effect of health, includes all the effects of health transmitted across generations. For example, genetic factors, parental dietary and hygienic habits and practices greatly affect children's health. When child survival rates increase, parents are more willing to substitute quantity of children with quality, giving fewer births to achieve the targeted number of children, and reducing the health burden of mothers and the financial and medical burden of the family. Meanwhile, the economy experiences demographic transition as a result of this quantity/quality trade-off with regard to children. Previous studies have discussed the above channels (e.g., Bloom and Canning (2000), Chakraborty (2004), Soares (2005), Doepke (2004), Galor (2005), Weil (2007), Jayachandran and Lleras-Muney (2009)).

Knowing that health influences individuals and countries through these three channels, one would be curious to learn the contributions of health on individual and national income, as well as economic growth. This paper examines the contribution of health to explain the differences in income levels across countries. In consonance with Shastri and Weil (2003) and Weil (2007), health is a form of human capital which serves as an input in goods production and affects labor productivity. (Hereafter, I term human capital in the form of health as "health-based human capital" or "health capital," interchangeably.) Instead of measuring the size of the health capital that enhances labor productivity directly,² I designed a method to measure *total health capital per worker*, and *the contribution of health capital to human capital* ("health capital share" hereafter). More specifically, to measure health capital per worker and health capital share, I construct an overlapping-generations model in which three mechanisms of health are embedded. The first of these mechanisms states that improvements in health capital increase labor productivity. The

²See the related literature for a comparison of Shastri and Weil (2003), Weil (2007) and the current paper.

second is that better health brings higher marginal product of physical capital and induces more capital accumulation. The third mechanism centers on altruistic parents making health investments for their children, which dictates the level of health capital of each child. The level of health capital subsequently determines the child's survival probability. Moreover, effectiveness of the parental health investment is impacted by health technology, or efficiency in the health sector. Based on the survival probability function, and with the use of survival probability and health expenditure data, estimates of health capital across countries are presented. To obtain the health capital share, the model is then calibrated using the estimation results and the data of the US economy. Finally, the relative total factor productivity (TFP) differences are calculated and compared to the relative TFP gaps obtained under the Solow and the Lucas production functions, in which only physical capital, or both physical and knowledge capital without externalities, are used as inputs in production.

This paper reveals that the introduction of health as an input factor greatly reduces the TFP gaps and logically explains cross-country income variations. For a 36-fold difference in output per worker, the model requires a 4.45-fold difference in the TFP. This compares with the 7.13 and 5.6-fold differences in the TFP gap for the Solow and the Lucas production functions, respectively. If we evaluate the performance of the model by examining the ability of the model to explain the variance of the output per worker, I find that physical capital, education-based human capital and health capital account for 23%, 11% and 8% of the cross-country income variances, respectively. Health plays an important role in explaining the income inequality across countries, though physical capital and education-based human capital have a greater effect. Moreover, the results also show that the poorer the country is, the more important the role of health and education becomes, especially for those very poor countries in sub-Saharan Africa and South Asia.

The remainder of this paper is organized as follows: Section 2 provides the basic framework, Section 3 deals with the optimization and the equilibrium of the model, and the numerical analysis and the calculations for the TFP gaps are relegated to Section 4. Finally, Section 5 concludes the paper.

Related Literature

When explaining cross-country income differences, researchers in development accounting usually investigate the differences in the level of productive factors across countries, or deal with efficiency in the production sectors or barriers in the economy. This paper tackles both, but with an emphasis on differences in productive factor levels. McGrattan and Schmitz (1999) and Caselli (2005) provide thorough surveys of the development accounting literature.

This paper closely relates to Shastry and Weil (2003) and Weil (2007) in the literature of development accounting. To determine the direct effect of health differences on income differences across countries, Shastry and Weil (2003) and Weil (2007) utilized: (1) microeconomic estimates of the direct effect of health on individual income; (2) aggregate data on health differences across

countries; and (3) a linear assumption between a scalar latent health measure and labor income”. This paper, on the contrary, relies on a survival probability function nested in a structural model to estimate the health capital across countries. Despite using a different approach, the main results are the same, i.e. health explains 9.9 and 8 percent of the log GDP per worker variance in Weil (2007) and the current paper, respectively.

Manuelli and Seshadri (2009) extended the Barro-Becker framework into a full lifecycle continuous time model with endogenous lifespan, human capital and fertility choice. The main departure of the current paper from their work is that health capital increases labor productivity; health capital in their work determines lifespan only. Their result reveals a large elasticity of TFP with respect to output per worker compared to that shown in this paper. In order to quantify the importance of human capital in the form of knowledge versus TFP in explaining income variations across countries, Erosa et al. (2010) adopt a heterogeneous individual model where agents are able to make choices in schooling quantity and quality. They find that a 20-fold difference in output per worker requires a 5-fold difference in TFP.

Instead of examining factor inputs to explain for cross-country income variations, Hall and Jones (1999), Prescott (1998), Parente and Prescott (2000), Cordoba and Ripoll (2008), and Restuccia et al. (2008) investigate the TFP, or the efficiency in production, and the barriers in the economic sectors. Parente and Prescott (2000) argue that barriers to adopting efficient technology in industries are the main reasons why poor countries remain mired in poverty. Chari et al. (1996) discuss inefficient government production. Cordoba and Ripoll (2008) adopt a framework of endogenous TFP, while Restuccia et al. (2008) focus on the barriers in the agricultural sector to explain for cross-country income differences.

Acemoglu and Johnson (2007), and Ashraf et al. (2008) investigated the relationship between health, income levels and development. They found that improvement in health brings higher fertility rates, which dilute the capital per worker and increase the dependent ratio. Conversely, Bloom et al (2004) claim that health has a positive and statistically significant effect on economic growth. Murphy and Topel (2005) measured the value of mortality decline and benefits of better health for the US. Other studies examining the relationship between health, labor productivity and development include Lorentzen et al (2005), Cutler et al. (2006), Aghion et al. (2010), Fogel (2004), Deaton and Arora (2009), Miguel and Kremer (2004), Wildman (2003), and Deaton (2003).

2 The model

The traditional human capital literature emphasizes the importance of education and work experience. However, as argued by Weil (2007) and Shastri and Weil (2003), such a formulation misses an important part of human capital – the health status of an individual. In line with their view,

this paper incorporates health as a component of human capital. In contrast with their empirical approach, the current work rests upon an overlapping-generations model of altruistic parents who care about the lifetime utility of their children. To focus on the role of health, I construct a model in which parents make health investments for their children, whereas the education level of the children is determined exogenously. Each generation consists of identical agents with perfect foresight. Each agent can live for at most three periods: childhood, young adulthood and old adulthood. Upon receiving health investments provided by their parents and being equipped with an exogenous level of education, children experience a mortality shock upon entering their young adulthood. Once they survive and become young adults, they immediately give birth to the new generation. They then work, consume, invest in the new generation's health and save for retirement. As they become old adults, they retire from their jobs, consume their savings, and exit the market at the end of their lives.

2.1 Population

Index time by t . In each period t , the economy is populated by three cohorts of agents: children, young adults and old adults. Denote the number of young adults in period t as N_t . To abstract from the fertility issue, I assume that the fertility rate is exogenous, and that each agent gives birth to f children right after she becomes a young adult. However, not every child can survive into young adulthood. A child experiences a mortality shock right before she enters into young adulthood,³ and the magnitude of the mortality shock depends on the health capital of the children. Denote health capital by h . For a child born in period $t - 1$ and equipped with health capital h_t when she turns to become a young adult, the survival probability is

$$\sigma(h_t) = \Phi(h_t; \mu, \varepsilon^2) \tag{1}$$

where $\Phi(h; \mu, \varepsilon^2)$ is the cumulative distribution function of the normal distribution with mean μ and variance ε^2 . Hence, if the current number of young cohorts is N_t , and supposing that every child is equipped with the same level of health capital (h_{t+1}), the population of young cohorts in the next period, N_{t+1} , is equal to $N_t f \sigma(h_{t+1})$.

2.2 Production technologies

There exists a continuum of competitive firms of measure one in this economy. In each period, a representative firm hires physical capital (K_t) and effective labor (L_t) to produce the single final good, which can be used for consumption as well as capital accumulation. The firm takes the

³The current study investigates how health affects labor productivity and parental health investments in children. Therefore, survival probabilities in adulthood, especially old-age survival problems, are not discussed in this paper.

competitive rental rate of physical capital r_t and the effective wage rate w_t as given, and operates using the following Cobb-Douglas technology:

$$Y_t = AK_t^\alpha L_t^{1-\alpha} \quad (2)$$

where $A > 0$ is the total factor productivity (TFP) and $\alpha \in (0, 1)$ denotes the physical capital income share. The effective labor is measured by $L_t = N_t H_t$, where N_t is the number of young agents and H_t is the human capital owned by each young worker. Human capital is assumed to be composed of health capital (h_t) and knowledge capital (m). The formation of human capital is assumed to take a Cobb-Douglas form with constant shares of health and knowledge,

$$H_t = h_t^\beta m^{1-\beta} \quad (3)$$

where $\beta \in (0, 1)$ measures the importance of health relative to knowledge in forming human capital: the bigger the β is, the more crucial the role of health is. The knowledge level, m , is assumed to be determined exogenously⁴. Health capital, h_t , is assumed to evolve according to

$$h_t = \xi x_{t-1}^\eta h_{t-1}^{1-\eta} \quad (4)$$

where x_{t-1} is the health investment made by parents, h_{t-1} is the health capital of the parents, $\xi > 0$ is the technology scaling factor in producing health, and $\eta \in (0, 1)$ denotes the relative importance of health investment to parental health transmission. That is, if the illness arising from genetic defects can be cured or alleviated by modern medical treatment, one will expect a more sizable η .

2.3 Preferences and budget constraints

Each agent can live at most for three periods: childhood, young adulthood and retirement. Consider a representative agent born in $t - 1$. As a child, she is entirely passive, receiving health investments offered by her parent and education exogenously provided. If she survives the mortality shock at the end of her childhood, she becomes a young adult and gives birth to f children. As a young adult, she is embodied with health capital h_t , knowledge capital m , and human capital $H_t = h_t^\beta m^{1-\beta}$. Meanwhile, she is endowed with a unit of labor time, which she supplies inelastically to the labor market at an effective wage rate w_t . With the wage earnings, she consumes (c_t^y), saves for her retirement (s_t), and decides the amount of health investments for her children. Assume that she equally favors her children and makes the same health investment (x_t) in each child. Her budget constraint when young is thus given by

$$c_t^y + s_t + f x_t = w_t h_t^\beta m^{1-\beta}. \quad (5)$$

⁴One can think of m as the knowledge level provided by public education and financed by per capita taxes.

As she steps into her old adulthood, she retires and consumes what she has saved when young, leaving no pecuniary bequest to her children:

$$c_{t+1}^o = R_{t+1} s_t, \quad (6)$$

where c_{t+1}^o stands for consumption when old and R_{t+1} is the gross rate of return on her savings. The representative young agent cares about her consumption when young and when old, as well as the lifetime consumption levels of all her surviving children.⁵ Assume that she values her surviving children with an intergenerational discounting factor (altruistic factor hereafter) $\gamma \in (0, 1)$. With log periodic utility, the optimization problem of the representative woman, born in period $t - 1$, can be specified as follows:

$$\begin{aligned} U_t = & \max_{c_t^y, c_{t+1}^o, s_t, x_t, h_{t+1}} \ln c_t^y + \frac{1}{1+\rho} \ln c_{t+1}^o + \gamma f\sigma(h_{t+1}) U_{t+1} \\ \text{s.t.} & \quad (5), (6), \text{ and } (4) \end{aligned} \quad (\text{P})$$

where $\rho > 0$ is the periodic discount rate.

The law of motion of physical capital in this economy is given by

$$K_{t+1} = (1 - \delta) K_t + Y_t - N_t c_t^y - N_{t-1} c_t^o - N_t f x_t \quad (7)$$

where $\delta > 0$ is the physical capital depreciation rate. That is, the stock of physical capital in the next period, K_{t+1} , is the sum of undepreciated physical capital, $(1 - \delta) K_t$, and unconsumed output, $Y_t - N_t c_t^y - N_{t-1} c_t^o - N_t f x_t$.

Now I am ready to proceed with solving the representative woman's problem and the equilibrium.

3 Optimization and Equilibrium

In this section, I focus only on the interior case where parents make a positive health investment. In the next section, I will calibrate the model to fit the U.S. economy. Using the calibration result of the U.S. economy, the analysis of cross-country income differences will then be performed.

3.1 Optimization

To solve the problem of the generation $t - 1$ representative woman, the first-order conditions with respect to c_t^y , c_{t+1}^o , s_t , x_t , h_{t+1} are derived. The two sets of first-order conditions are given by:

$$\frac{1}{c_t^y} = \frac{R_{t+1}}{(1 + \rho) c_{t+1}^o} \quad (8)$$

⁵I assume that the agent derives zero utility from her children if her children pass away before they become young adults.

$$\begin{aligned} & \frac{f x_t^{1-\eta} h_t^{1-\eta}}{c_t^y \eta \xi} \\ = & \gamma f \sigma'(h_{t+1}) U_{t+1} + \frac{\gamma f \sigma(h_{t+1}) \beta w_{t+1} h_{t+1}^{\beta-1} m^{1-\beta}}{c_t^y} + \frac{\gamma f \sigma(h_{t+1})}{c_t^y} \frac{f x_{t+1}}{h_{t+1}} \frac{1-\eta}{\eta} \end{aligned} \quad (9)$$

Equation (8) is the intertemporal Euler equation, stating that the marginal utility of consumption when young is equal to the marginal cost of postponing consumption when young, which is the total marginal utility of consumption when old that the agent can enjoy if she postpones her consumption. Equation (9) gives the *intergenerational trade-off of health investment*. The left-hand side (LHS) of (9) is the marginal cost of increasing one unit of health capital for all the children in terms of forgone consumption; the first term on the right-hand side (RHS) stands for the utility the mother derives directly from her children when the survival probability rises as the children's health capital increases; the second term is the utility the parent enjoys directly as her children's wage earnings increase due to better health and hence higher labor productivity; the last term represents the marginal benefits from a higher health stock for the future generation – a higher health stock for the children makes future health investments more productive.

The optimization problem of the competitive firm implies that the rental rate of physical capital r_t and the effective labor wage rate w_t are

$$r_t = A \alpha K_t^{\alpha-1} L_t^{1-\alpha} \quad (10)$$

$$w_t = A (1-\alpha) K_t^\alpha L_t^{-\alpha} \quad (11)$$

Therefore, the gross rate of return received by the old adults is equal to

$$R_t = 1 - \delta + r_t \quad (12)$$

3.2 Equilibrium

All the discussions below only focus on the steady-state equilibrium. To simplify the analysis, the exogenous fertility rate f is assumed to be equal to the inverse of the survival rate of children $\sigma(h)$. Assume that the number of the first young cohorts equals N . In this manner, the population of children in each period is equal to Nf , and the numbers of both young and old cohorts are constant and equal to N over time.

Definition 1: *Given a knowledge level m , a steady-state equilibrium in this economy is a tuple of quantities $\{H_t, h_t, c_t^o, c_t^y, s_t, x_t, K_t, Y_t\}_{t=0}^\infty$ together with a pair of prices $\{w_t, r_t\}_{t=0}^\infty$ such that:*

- (i) *Each woman optimizes;*
- (ii) *Firms maximize profits;*

- (iii) *Markets for goods, loanable funds and labor all clear;*
- (iv) *The population of the young and old cohorts remains unchanged for a given fertility rate f ;*
- (v) *All the endogenous variables are constant.*

Define $y \equiv \frac{Y}{N}$ and $k \equiv \frac{K}{N}$ as output per worker and physical capital per worker, respectively. In the steady state equilibrium, output per worker, the physical capital rental rate and the effective wage rate can be rewritten as $y = Ak^\alpha (h^\beta m^{1-\beta})^{1-\alpha}$, $r = A\alpha K^{\alpha-1} (NH)^{1-\alpha} = A\alpha k^{\alpha-1} (h^\beta m^{1-\beta})^{1-\alpha} = r(k, h)$ and $w = A(1-\alpha) K^\alpha (NH)^{-\alpha} = A(1-\alpha) k^\alpha (h^\beta m^{1-\beta})^{-\alpha} = w(k, h)$, respectively. From the health capital evolution equation (4), steady-state health capital is proportional to the health investment:

$$h = \xi^{\frac{1}{\eta}} x \quad (13)$$

Now I make use of the individual budget constraints both when young and when old. From (8), $c^o = \frac{Rc^y}{1+\rho}$ and $s = \frac{c^o}{R} = \frac{c^y}{1+\rho}$ follow the budget constraint when old (6). By substituting $x = \xi^{\frac{-1}{\eta}} h$ and $s = \frac{c^y}{1+\rho}$ into (5), c^y is solved as

$$\begin{aligned} c^y &= \frac{1+\rho}{2+\rho} \left[A(1-\alpha) k^\alpha (h^\beta m^{1-\beta})^{1-\alpha} - \xi^{\frac{-1}{\eta}} h \right] \\ &= c(k, h) \end{aligned}$$

That is, c^y is a function of k and h . Therefore, c^o and s are functions of k and h , and can be written as $c^o(k, h) = \frac{[1-\delta+r(k, h)]c(k, h)}{1+\rho}$ and $s = \frac{c(k, h)}{1+\rho}$. In the steady state, equation (7) can hence be re-expressed as an equation in (k, h) :

$$\delta k = Ak^\alpha (h^\beta m^{1-\beta})^{1-\alpha} - \left[1 + \frac{1-\delta+r(k, h)}{1+\rho} \right] c(k, h) - \xi^{\frac{-1}{\eta}} h \quad (14)$$

In the steady state, since the health capital across generations are constant, the survival probability is also constant. The consumption for the young cohorts and the old cohorts are the same across all generations, respectively. Therefore, the lifetime utility can be obtained as $U = \frac{1}{1-\gamma f \sigma(h)} \left[\ln c^y + \frac{1}{1+\rho} \ln c^o \right]$. By using $h = \xi^{\frac{1}{\eta}} x$, equation (9) can be rearranged as

$$\frac{\xi^{\frac{-1}{\eta}}}{\eta} [1 - \gamma f \sigma(h) (1 - \eta)] = \frac{\gamma \sigma'(h) c(k, h)}{1 - \gamma f \sigma(h)} \left[\ln c(k, h) + \frac{\ln c^o(k, h)}{1 + \rho} \right] + \gamma \sigma(h) \beta A (1 - \alpha) k^\alpha h^{\beta(1-\alpha)-1} m^{(1-\beta)(1-\alpha)} \quad (15)$$

which is also an equation in (k, h) . Equations (14) and (15) are therefore the two nonlinear equations in (k, h) , governing the equilibrium of this economy.

It is impossible to have an analytical solution for k and h , but one can still gain some intuition by examining the two nonlinear equations. Consider first that parents become more altruistic as γ increases. Increases in γ affect (15) only and leave equation (14) unchanged. Suppose $\sigma'(h) > 0$ and

$\sigma''(h) < 0$ and consider the case where $c(k, h)$ is increasing in k and h , in response to an increase in γ and to restore the balance in (15), the equilibrium value of h is more likely to rise. As h increases, the marginal product of physical capital increases, and hence the equilibrium value of k is more likely to rise as well. Next, consider a shift in the survival probability distribution as illustrated in Figure 4 with the initial steady-state health capital given at h_0 . A shift in the survival probability not only improves the current survival probability but also increases the marginal product of health capital in reducing mortality. With the same health expenditure, the number of children surviving increases. Parents now enjoy more utility from surviving children and may wish to increase their own consumption by reducing their health expenditure on children. Alternatively, parents may want to invest more in the health of their children as the marginal product of health capital is now higher. Hence, the direction of changes in the equilibrium health capital is ambiguous. Now consider advancements in the health technology scaling factor ξ . From (14), as ξ increases, resources allocated to health accumulation can be reduced, and one can expect that there will be more resources available for consumption and physical capital accumulation. Now examine equation (15) and consider the case where $c(k, h)$ is increasing in (k, h) . Since $\eta \in (0, 1)$, $\gamma \in (0, 1)$ and $f\sigma(h) = 1$, from (15) $\gamma f\sigma(h)(1 - \eta) < 1$, when ξ increases, with $\sigma'(h) > 0$ and $\sigma''(h) < 0$, one would expect h to increase. From (14), it is more straightforward that h increases as ξ increases. Therefore, it is more likely that the equilibrium h increases as the health technology improves. The equilibrium value of k is more likely to rise as h increases due to a higher marginal product of physical capital, while the magnitude of the increase in k is much smaller than the magnitude of the increase in h .

Although no analytical solution exists, equation (14) and (15) can still be plotted in the (k, h) -plane for illustration, as shown in Figure 5. Name equation (14) “*the capital law of motion locus* (KL)” and equation (15) “*the intergenerational trade-off locus* (IT).” the equilibrium results of the shifts in the parameters corroborate the discussion in the last paragraph under the parameterization of the U.S. economy in the next section. Since health and physical capital are complements in production, the KL locus, describing the resources available for physical capital and health capital accumulation, is upward sloping. On the contrary, the IT locus, depicting the parental trade-off for old-age consumption (financed by savings, which determines the physical capital level in the future) and children’s health investment, is downward sloping. Given the wage income, parents can never simultaneously save and increase their investment in children’s health. Figure 5 shows that, under the parameterization of the U.S. economy, the IT locus shifts up as parents become more altruistic ($\gamma \uparrow$) and the equilibrium values of k and h both increase. As health technology becomes more effective ($\xi \uparrow$), both the KL and the IT loci shift upward, leading to an increase in the equilibrium values of both k and h .

Since the focus of this paper is on studying cross-country income differences, and examining whether health serves as a plausible factor in explaining international income differences, I shall

proceed to calibrate the model to fit the U.S. economy and investigate the role played by health in a cross-country setup.

4 Numerical Analysis

In this section, using cross-country data, I first estimate the parameters of the survival probability function and the health capital production function. The results of the estimation are then taken to calibrate the model to fit the U.S. economy. With the calibrated parameters for the production technologies and the estimates of health capital, I compute the TFP differences required to explain the cross-country income variations predicted by the model. Comparisons of the TFP differences between the current model, the neoclassical growth model (with physical capital as the only input in production), the Lucas-type model (with physical capital and education as inputs in production but without externalities), and the model where the health technology are the same across countries are presented. I also provide the variance decomposition to examine the performance of the model according to Caselli (2005).

4.1 Estimation and Calibration

The key question the current study asks is: based upon the goods production technology given by (2), how much can the differentiations in health explain the cross-country income differences as well as the required TFP differences to match the output performance shown in the data? Thus, the purpose of the estimation and calibration exercises is to pin down the parameters, and to find the estimates needed to throw into production function (2) to conduct the cross-country comparisons.

Before plunging directly into the estimation and the calibration, let us examine what is needed to carry out the cross-country analysis. All the analyses are conducted assuming that the economies under study are in the steady state. As shown in the previous section, by using (2) and (3), one can derive the output per worker formula which is given as follows:

$$y = Ak^\alpha \left(h^\beta m^{1-\beta} \right)^{1-\alpha} \quad (16)$$

The physical capital per worker k can be imputed from the Penn World Table (PWT); estimates of the capital income share α of the U.S. mostly fall within the range (0.25, 0.4).⁶ There are also different measures of knowledge capital which can be plugged into the calculation here. Caselli (2005) summarizes and experiments with different measures of human capital (knowledge-based human capital in this paper), including different measures for years of schooling, working experiences, and

⁶Capital income shares vary across countries (e.g., Gollin (2002)), but most estimates fall within the range of (0.25, 0.4). In the literature on development accounting, quantitative analyses are usually performed with a reasonable choice of capital income share under the assumption that the income share is the same across countries.

quality of schooling (teachers' human capital, pupil-teacher ratios and test scores), and finds that these different measures do not alter the results of the explanatory power of knowledge capital in the cross-country differences in income greatly. Thus, I will compute the knowledge capital based upon the Mincerian rate of return on the years of schooling only. However, there is no estimate of health capital h and the health capital share of human capital β that can be readily applied to the current study.⁷ Therefore, the remaining task is to figure out a way of obtaining the health stock across countries and the health capital share.

4.1.1 Estimation

Health capital is estimated through the use of the survival probability function. From (13), health capital is linear in health expenditure with the coefficient $\xi^{\frac{1}{\eta}}$. Note that the health technology scaling factor ξ , which is unobservable, captures not only the health advancements of a country, such as R&D in medicine, treatments, and medical practice procedures, but also the environmental factors that could affect the efficiency of health investment. Examples include general sanitation conditions, pollution, malpractices, and bureaucracy within the health system. Available data from the World Health Organization (WHO) to approximate the health technology across countries include density of healthcare infrastructure (including health centers, district or rural hospitals, provincial hospitals and regional hospitals), density of health management and support workers, density of physicians, density of nurses and midwives, density of hospital beds, improved sanitation facilities access rates, and improved water source access rates. When choosing the data to approximate for the health technology across countries, I adopt two criteria: availability and appropriateness. Although densities of physicians, nurses and midwives as well as hospital beds are believed to be measures of easy access to medical services, the training of the medical personnel in developing countries are worse than that of developed countries, and medical practitioners are found to be absent from their work in developing countries. Hospitals in less developed nations are found to be poorly managed compared to advanced economies, with drugs stored incorrectly, hospital equipments maintained improperly and resulting in high breakdown rates.⁸ Also, the data for density of health infrastructure are only available for developing countries, preventing one from using the data for conducting a larger-sample of cross country studies. Furthermore, a simple check on the survival rates at age 25 and the available data reveals that the improved sanitation facilities access rate is the most highly correlated variable.⁹ Therefore, I choose to use the improved sanitation facilities coverage rate to

⁷Murphy and Topel (2006) provide estimates of the health capital of the U.S. based on the willingness to pay for health, which is very different from the current study. Shastry and Weil (2003) and Weil (2007) estimate the portion of health capital that contributes directly to the labor productivity, which is also different from this paper.

⁸See Hossain and Osman (2007) and Nafula et al. (2004).

⁹The correlations between the survival rate at age 25 and the density of physicians (per 1,000), the density of nurses and midwives (per 1,000), the density of hospital beds (per 1,000 in 1990), the improved sanitation facilities

approximate for the health technology scaling factor. The data for improved sanitation facilities access rate used are for the year 2000 when available, and the closest year is used if data for the year 2000 are not available. Countries that do not have data are dropped, except for Italy and New Zealand. Italy and New Zealand are both advanced economies, and therefore I regard them as having a 100 percent improved sanitation facilities coverage rate as this is the case for all other advanced economies. Denote the improved sanitation facilities coverage rate of country i as θ_i . The health technology scaling factor ξ_i of country i is assumed to follow $\xi_i = b\theta_i$, where b is a scaling parameter proxying the magnitude of ξ_i . The data on health expenditure per worker are calculated using health expenditure as a percentage of total GDP during 1995-2000, obtained from the World Bank. To calculate the health expenditure per worker, I first calculate the number of workers from the PWT 6.3, by dividing total real GDP (rgdpl $\times$ POP) by real GDP per worker (rgdpwok). Total health expenditures are computed by multiplying health expenditure as a percentage of GDP to total real GDP. I then divide total health expenditures by the number of workers to obtain health expenditure per worker. After that, I normalize health expenditures with the normalization that the real GDP per worker of the U.S. is equal to 1 in the year 2000. Finally, I take the geometric average of the health expenditure per worker for 1995-2000 to arrive at the health expenditure per worker used in the paper. For the survival probability, I choose to use the survival rate at age 25 so as to match the definition of the survival probability in the model.¹⁰ The data for the survival probability at age 25 are also obtained from the WHO and computed by dividing l_x at age 25 (the number of people surviving at age 25 out of 100,000 people) by 100,000 (live births) in the life table compiled by the WHO. The data are for the year 2000 and are for both sexes.

We are ready to estimate the following equation:

$$\begin{aligned}\sigma_i(h_i) &= \Phi(h_i; \mu, \varepsilon^2) \\ &= \Phi((b\theta_i)^{\frac{1}{\eta}} x_i; \mu, \varepsilon^2)\end{aligned}$$

where σ_i is the survival rate at age 25 for country i . With data for improved sanitation facilities access rate and health expenditure, one can only estimate two parameters. Therefore, the variance of the normal distribution is normalized to one and the health investment share η is assumed to be 0.5, i.e. health investment and parental health are equally important for children's health.¹¹ The results of the nonlinear least squares estimation are: $b = 10.8470$ and $\mu = -0.9241$. Table 2 reports the data for the survival probability, the survival probability predicted by the model, and the improved sanitation facilities coverage rate by income decile. Though not perfect, the predicted results of access rate, and the improved water source access rate are 0.7263, 0.5499, 0.5265, 0.8642 and 0.8188, respectively.

¹⁰The survival rate at age 25 is defined as the number of people still alive at age 25 out of 10,000 live births. In the model, a mortality shock strikes agents at the end of the first period of their lives. Assume that each period represents 25 years. Hence, the survival rate at age 25 provides the best fit to the model.

¹¹The likelihood under different choices of the health investment share is almost the same.

the model match the pattern of survival probabilities across countries in general. Since more than 70 percent of the countries under examination have a survival probability greater than 0.9, the predicted survival probability is truncated at 0.82, which is higher than the survival probabilities for countries in the bottom 20th decile. In the next subsection I will turn to calibrating the U.S. economy using the estimation results obtained here.

4.1.2 Calibration

Other than η , ξ , μ and ε , the remaining parameters and exogenous variables of the model are: (i) preference parameters – time preference rate (ρ) and altruistic factor (γ); (ii) parameters related to production and resources – TFP (A), income share of physical capital (α), physical capital depreciation rate (δ), and the health capital share in forming human capital (β); and (iii) the exogenous variables – fertility rate (f) and knowledge capital (m).

To calibrate for the remaining parameters in the model, I choose the U.S. economy as the calibration benchmark. Since the agents can live for at most three periods in the model, one model period is set as 25 years. When computing the health expenditure per worker, the real GDP per worker of the U.S. in the year 2000 is normalized to one. Hence, the output per worker per model period corresponds to 25. The exogenous fertility rate f for the U.S. is set at 1.0187, which is equal to the inverse of the survival rate at age 25 for the U.S. in the year 2000. The implied fertility rate per woman for this chosen fertility rate for the U.S. is roughly 2.04, which is very close to the U.S. fertility data.¹² The 25-year time preference rate is set at $\rho = 2.3864$, and corresponds to an annual time preference rate of 5%. The physical capital share of income is set at $\alpha = 0.36$, as is commonly used in the macroeconomics literature. The investment-to-output of the U.S. is set at 0.2, roughly equal to the average investment-to-output ratio of the U.S. during 1970-2000. The depreciation rate δ is set at 2.0067 to match the investment-to-output ratio of 0.2.¹³

The parameters remaining are γ , A and β . With two equations (14) and (15), I can pin down two parameters only. I assume that the “true” altruistic factor is 0.8, meaning that parents slightly discount the children’s lifetime utility compared to their own consumption. After adjusting for the time discount factor, the altruistic factor is set at $\gamma = 0.2362$.¹⁴ The investment-to-output ratio for

¹²The total fertility rate of the US in 2000 was 2.06.

¹³Following Caselli (2005), the annual depreciation rate is set at 0.06 to calculate the physical capital per worker from the PWT 6.3. The 25-year depreciation is thus equal to 0.7871, meaning that the value of a machine is roughly equal to 21% of its original value after 25 years of use. However, in reality a machine may be outdated in 10 years. Therefore, instead of setting the 25-year depreciation rate at 0.7871, the 25-year depreciation rate is set at 2.0067 to match the investment-output ratio of 0.2. This number means that a factory needs to replace its machinery roughly twice in every 25 years.

¹⁴There is no general consensus on the magnitude of the altruistic factor. Manuelli and Seshadri (2009) adopt a continuous time framework and calibrate the parental altruism to match the fertility rate. They use two parameters

the U.S. is 0.2. The physical capital per worker for the U.S. in 2000 is 2.49, as computed from the PWT 6.3.¹⁵ The health capital of the U.S. is 14.72 from the estimation in the previous subsection. To measure knowledge capital, a constant rate of return to schooling is used: $m = \exp(\phi(s_y)) = \exp(\phi_s s_y)$. s_y is the average years of schooling of people aged 25 and over in the U.S. in the year 2000 taken from Barro and Lee (2010) and ϕ_s is the Mincerian return on schooling, which I set at 10% as reported in Psacharopoulos and Patrinos (2004).¹⁶ Note that from (12), A can be rewritten as a function of the annual interest rate and parameters α , β and δ . Let r_a denote the annual interest rate. The gross return R is equal to $(1 + r_a)^{25}$. By using (12), A can be written as $A = \frac{(1+r_a)^{25+\delta-1}}{\alpha k^{\alpha-1} (h^\beta m^{1-\beta})^{1-\alpha}} = A(r_a; \alpha, \beta)$. Now divide (14) by y and re-express (14) in terms of expenditure shares to total output. In this way, consumption when young c^y and when old c^o both depend on the annual interest rate. Since total output per period is 25, the annual interest rate can be solved directly from (16), which is equal to 0.03905. Finally, β can be pinned down from (15), and is equal to 0.0595. Table 3 summarizes the targets and assumptions used along with the results of the estimation and calibration.

4.2 Cross-country analysis – general result

Now we are ready to perform the cross-country analysis. Using the same method for calculating the capital per worker for the U.S., the capital per worker across countries is calculated using data from PWT 6.3¹⁷. Data on years of schooling for people aged 25 and over in 2000 are taken from Barro-Lee Educational Attainment Dataset (2010). To measure knowledge capital, in line with the approach to measuring the knowledge capital of the U.S., a constant Mincerian return on years of schooling is used. Psacharopoulos and Patrinos (2004) provide the latest estimates of the Mincerian return on years of schooling for 98 countries. For countries that do not have available estimates, to capture parental altruism, with an altruistic factor equal to $\exp(-\alpha_0 + \alpha_1 f)$. They match their altruistic factor to the fertility rate, and obtain $\alpha_0 = 0.24$ and $\alpha_1 = 0.65$. According to their argument, a parent is completely altruistic if α_0 is equal to zero and α_1 is one. Hence, to make their altruistic factor comparable to the framework adopted here, I compute the "true" altruistic factor as $\frac{e^{-\alpha_0 + \alpha_1 f}}{e^f} = 0.77$, which is smaller than one. Jones and Schoonbroodt (2011) assume that the altruistic factor is equal to one. Doepke and Zilibotti (2005) set the altruistic factor to one when simulating their model. In their 2008 work, they use an altruistic factor of 0.53 in their numerical example.

¹⁵Following the procedure outlined in McGrattan and Schmitz (1998), I first compute the capital-output ratio of the U.S. for the period 1970-2000 using PWT 6.3 data. I then multiply the capital-output ratio by total real GDP and divide it by the number of workers to obtain the capital per worker.

¹⁶Erosa et al. (2010) report similar estimates of Mincerian returns in the range of 0.08 to 0.12. They also use a Mincerian return of 0.1 for the U.S.

¹⁷When calculating the physical capital per worker, I take the investment share (ki), real GDP per worker (rgdpwok), population (POP), and real GDP per capita (rgdpl) data for the period from 1970 to 2000 from PWT 6.3. Countries that are established after 1970 or that have no data for most of the time are dropped.

I proceed with assigning them with estimates in the following manner. First, all the countries are grouped into different regions according to Barro and Lee (2010). These regions include Advanced Economies, Europe and Central Asia, South Asia, East Asia and the Pacific, Latin America and the Caribbean, sub-Saharan Africa, and the Middle East and North Africa. Then, group means are calculated and assigned to countries with no estimates of the Mincerian return on years of schooling.

Hall and Jones (1999) construct a piecewise linear function using the estimates of Mincerian wage regressions reported by Psacharopoulos (1994). For the first four years of schooling, the return to schooling is assumed to be 13.4 percent, which is the average for sub-Saharan Africa. For years of schooling between 4 and 8, the return to schooling is assumed to be 10.1 percent, which is the world average. For years of schooling beyond 8 years, the return is 6.8 percent, as this is the return for OECD countries. I see their method as an alternative to measuring knowledge capital. The sensitivity test using the Hall and Jones (1999) measure of knowledge capital shows that the main result of the paper does not change with this alternative measure of knowledge capital.

All the analyses conducted below are based on equation (16). Table 4 reports the data – the survival rate at age 25, the relative real GDP per worker, the average years of schooling of people aged 25 and over, the physical capital per worker, and the improved sanitation facilities coverage rates for 117 countries by income deciles. Richer countries have higher survival rates, longer years of schooling, and larger stocks of capital per worker. Before going directly to the key result, i.e., learning whether health plays an important role in counting cross-country income differences, I list the models to be compared as follows:

- Model K: the Solow model with physical capital as the only input in goods production.
- Model KM: the Lucas model with both physical and knowledge-based human capital without externalities as inputs in goods production.
- Model KMX: the model with physical capital, knowledge-based and health-based capital as inputs in producing goods, while all the countries have the same technology in the health sector.
- Model KMH: the model with physical, health and knowledge capital as inputs in producing goods.

Table 5 reports the key results of the paper. Model K and Model KM predict that the top 10th percentile countries are 7.13 and 5.60 times more effective in carrying out production than the bottom 10th percentile countries, respectively, while Model KMX and Model KMH predict that the top 10th percentile countries are only 4.92 and 4.45 times more productive than the bottom 10th percentile countries. In short, to explain a 35-fold difference in output per worker, when introducing knowledge-based human capital as production input, the required TFP differentials drop by roughly

21 percent. As health-based human capital is introduced into the model, if all countries are equally efficient in producing health capital, the required TFP gaps reduce by 12 percent. When the health technology is allowed to differ across countries, the required TFP gaps further fall by 8 percent. That is, when counting both the health technology efficiency and the level of health investments, the required TFP gaps decrease by 20 percent compared to Model KM.

Another way to evaluate the performance of the model is to examine the variance decomposition and calculate how much of the variance the input factors can capture. Following Caselli (2005), I compute the ratio of the variance of log income that contributes by input factors to the variance of log GDP per worker. The result is reported in Table 6. It is shown that physical capital can explain up to 23% of the variance in log GDP per worker. Adding education-based human capital can explain an additional 11-12% of the variance. When health capital is introduced as an input factor in the production, the model further captures 8% of the variance in log GDP per worker. These results indicate that, although ranked below physical capital and knowledge-based human capital, health capital is still a very important factor in explaining for cross country income differences.

The characteristics of the TFP, or the Solow residuals, refer to the residuals unexplained by factors or efficiency measures specified in the model. Hence, the TFP gaps calculated from an “ideal” model should not be too big. In examining Table 5, one can see that when going from Model K, Model KM, Model KMX to Model KMH, although the relative TFPs of countries under the 80th percentiles are converging to one, the relative TFPs of the top 20th percentile countries are increasing. This either signifies the inefficiency of the U.S. with its capital stock – the U.S. should have higher output with its large stock of knowledge-based and health-based capital, or the estimate of the knowledge capital and health capital of the U.S. are too high. The discussion of estimation of knowledge capital is dismissed since the role of knowledge-based human capital is not the focus of the current study. It is noted that the health expenditures in the U.S. have been increasing incessantly and the share of health expenditures to the total GDP for the U.S. is higher than that for the OECD countries. By calculating health expenditure as a percentage of GDP using data in Carter et al (2006), health expenditure as a percentage of total GDP of the U.S. has increased from 3.46 percent in 1929 to 13.5 percent in 2000. Thus, one has reasons to believe that the U.S. is less efficient in the health sector, and the health capital estimated based on the health expenditure should be adjusted. To reflect the “inflation” of the health capital in the benchmark estimation due to the inefficiency in the U.S. health sector, I assume that the health expenditure per worker for the U.S. is half of its original value.^{18,19} The relative TFPs are reported in the last column in Table 5. By comparing this experiment to Model KM and Model KMH, it is found that this experiment

¹⁸The average health expenditure per worker for countries with a real GDP per worker that is 80 percent of that of the U.S. or higher is 0.0641, which is roughly equal to half of the health expenditure per worker for the U.S.

¹⁹With this assumption, I am assuming that the health capital of advanced economies should not depart too much from each other.

is quite successful: the relative TFP of the top 10th percentile and the bottom 10th percentile countries are lower and higher or roughly equal than those in Model KM and the benchmark Model KMH, respectively, revealing a reduction in the variance in relative TFPs. Now compare the results of this experiment to Model K and Model KM – again, 37 percent and 20 percent reductions in required TFP differences from Model K and Model KM, respectively, which roughly maintain the same results as the benchmark KMH model.

Table 7 compares the data and the model predicted endogenous variables. Most of the countries under study have survival rates at age 25 of more than 0.9, and hence the model predicted survival rates at age 25 are higher than the data, especially for the bottom 20th percentile countries. As mentioned in the last paragraph, health expenditure in the U.S. is much higher than that of countries with income levels similar to those in the U.S. Thus, the estimated health capital based on health expenditure for the U.S. is much higher than that of other countries in the same development stage as the U.S. Since the U.S. economy is the calibration benchmark, it is not surprising that the model predicted health expenditure and health capital are much higher than those estimated in the data. For countries with lower income percentiles, an over-predicted survival rate at age 25 together with the parameters based on the U.S. economy translate into much higher model predicted health capital and health expenditures. Since higher levels of health capital lead to higher labor productivity and higher marginal product of physical capital, the model predicted physical capital is also higher than the data. As long as the U.S. economy is the calibration benchmark, the health capital and the health expenditure predicted by the model will exhibit the same pattern as that reported in Table 7. In spite of the over prediction of physical capital, health capital and health expenditure, the model predicted output per worker matches to the data quite well.

4.3 The poor nations

Economists have been requesting an answer as to why poor nations stay poor but have not been too successful in finding an answer up to now. In this subsection, I try to tackle this question. Since the countries of concern are poor nations, I group countries with a real GDP per worker of less than 10 percent of that of the U.S. by region: 36 nations are categorized as poor countries out of 117 countries. These poor nations are scattered in five regions: (1) East Asia and the Pacific: Cambodia, Laos, Vietnam, Mongolia, Papua New Guinea, China; (2) Europe and Central Asia: Albania; (3) Latin America and the Caribbean: Guyana, Nicaragua and Haiti; (4) South Asia: Bangladesh, Nepal, and India; and (5) sub-Saharan Africa: Benin, Burundi, Cameroon, Central African Republic, Democratic Republic of Congo, Cote d'Ivoire, Gambia, Ghana, Kenya, Lesotho, Malawi, Mali, Mauritania, Mozambique, Niger, Rwanda, Senegal, Sierra Leone, Sudan, Tanzania, Togo, Uganda, and Zambia. Table 8 reports the average real GDP per worker, the survival rate at age 25, the improved sanitation facilities coverage rate, average years of schooling, and the relative

TFP differences calculated from Model K, Model KM and Model KMH for these countries by region.²⁰

In general, when India is excluded, the average years of schooling in South Asia and sub-Saharan Africa are very close, but the health condition in South Asia is much better. China and India perform better in terms of output per worker, health and education, and Latin American and Caribbean countries also perform better than other regions. Nonetheless, the results in Table 8 still show that health has great explanatory power. To see this, I compare the relative TFPs of East Asian, South Asian and sub-Saharan countries. When moving from Model K to Model KM, the relative TFPs all increase. This indicates that educational attainment in these countries is low, and variations in physical capital cannot account for the large income differentials across countries. The introduction of human capital in the form of education hence can capture the residuals unexplained by physical capital. I then turn to compare Model KM with Model KMX. With the introduction of health capital with the health technology held at the same level across countries, the relative TFPs increase by roughly 0.0407-0.0475 for all low income countries except countries in Latin America and the Caribbean. When the health technology are allowed to be different across nations, the relative TFPs increase by roughly 0.0303-0.0436 for low income countries except countries in Latin America and the Caribbean. This result contrasts to the result in the previous subsection. For countries with higher income levels, the inclusion of health as an input in goods production does not affect the relative TFPs greatly. On the contrary, for poor countries, the introduction of health increases the relative TFPs significantly, with the differences in health efficiency contributing substantially. This phenomenon is especially notable in South Asia and sub-Saharan Africa, as the variations in the health efficiency and technology account for roughly half of the increases in the relative TFPs, and the variations in the level of health investment account for the remaining half of the increase in the relative TFPs.

I therefore conclude that the role of education or knowledge is important, the role of health is also responsible for the low income levels in poor countries, and the explanatory power of health for the low income levels is dependent upon the geographic region. The efficiency of producing health, referring to the adversity of the health environment, the true health technology a country owns, and the bureaucracy in the health service system, is as important as the health investment levels for poor countries, especially for South Asian and sub-Saharan countries.

5 Conclusion

This paper illustrates the significant role of health in explaining for income inequality across countries by examining both the efficiency and the factor inputs in the health sector in developing

²⁰ Albania is the only observation in the European area, and hence I do not report it in Table 8.

accounting. A poor health environment makes health investment less effective, translates into a higher mortality rate, affects the parents' attitude toward children, and retards human capital accumulation and output production. Based on a structural model, this paper proposes a completely new approach in estimating health-based human capital. The results show that, with the introduction of health as a factor input, the required TFP gap to explain a 35-fold difference in income is 4.45, which is much smaller than the required TFP gaps implied by the Solow and the Lucas production functions. The results further demonstrate that, while knowledge is crucial to explaining cross-country income differences, health is also important in addressing the income inequality across nations. For extremely poor nations in sub-Saharan Africa and South Asia, the efficiency of transforming health investments into good health is equally important as the amount of the health investment.

There are several possible extensions for this paper. One possible extension is to consider endogenous fertility and education choice. Recall that in the model, the steady-state fertility rate is assumed to be the inverse of the survival probability. Although well-suited for rich countries, this assumption understates the fertility rates in poor countries and possibly leads to the over prediction of physical capital and health capital. Other possible extensions include differentiating between public and private health expenditures and incorporating endogenous health technology diffusion. Finally, in addition to being used to study the issues in income and health variations across countries, my model can also be generalized into a heterogeneous agent framework to study the within country income and health inequality.

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Appendix

A.1 Countries

There are 117 countries under studied in this paper. The list of countries are provided below (countries are ordered alphabetically).

Albania, Algeria, Argentina, Australia, Austria, Bangladesh, Barbados, Belgium, Belize, Benin, Bolivia, Botswana, Brazil, Bulgaria, Burundi, Cambodia, Cameroon, Canada, Central African Republic, Chile, China, Colombia, Congo, Dem. Rep., Congo, Republic of, Costa Rica, Cote d'Ivoire, Cuba, Cyprus, Denmark, Dominican Republic, Ecuador, Egypt, El Salvador, Finland, France, Gabon, Gambia, Germany, Ghana, Greece, Guatemala, Guyana, Haiti, Honduras, Hungary, Iceland, India, Indonesia, Iran, Ireland, Israel, Italy, Jamaica, Japan, Jordan, Kenya, Korea, Kuwait, Laos, Lesotho, Libya, Luxembourg, Malawi, Malaysia, Maldives, Mali, Malta, Mauritania, Mauritius, Mexico, Mongolia, Morocco, Mozambique, Namibia, Nepal, Netherlands, New Zealand, Nicaragua, Niger, Norway, Pakistan, Panama, Papua New Guinea, Paraguay, Peru, Philippines, Poland, Portugal, Qatar, Rwanda, Saudi Arabia, Senegal, Sierra Leone, Singapore, South Africa, Spain, Sri Lanka, Sudan, Swaziland, Sweden, Switzerland, Syria, Tanzania, Thailand, Togo, Trinidad & Tobago, Tunisia, Turkey, Uganda, United Arab Emirates, United Kingdom, United States, Uruguay, Venezuela, Vietnam, Zambia, Zimbabwe

A.2 Derivation of the model

To solve for the generation- t (born in $t-1$) representative agent's problem, I denote λ_{1t} , λ_{2t} , and λ_{3t} as the present value Lagrange multipliers associated with (5), (6), and (4). The first-order conditions with respect to c_t^y , c_{t+1}^o , s_t , x_t and h_{t+1} are:

$$\frac{1}{c_t^y} = \lambda_{1t} \quad (17)$$

$$\frac{1}{(1 + \rho) c_{t+1}^o} = \lambda_{2t} \quad (18)$$

$$\lambda_{1t} = \lambda_{2t} R_{t+1} \quad (19)$$

$$\lambda_{1t} f = \lambda_{3t} \eta \xi x_t^{\eta-1} h_{t-1}^{1-\eta} \quad (20)$$

$$\gamma f \sigma'(h_{t+1}) U_{t+1} + \lambda_{1t+1} \beta w_{t+1} h_{t+1}^{\beta-1} m^{1-\beta} + \lambda_{3t+1} (1 - \eta) \xi x_{t+1}^{\eta} h_{t+1}^{-\eta} = \lambda_{3t} \quad (21)$$

Since the Lagrange multipliers are in present value, one needs to derive the first-order condition with respect to c_{t+1}^y to derive the relationship between λ_{1t} and λ_{1t+1} , which is given by:

$$\frac{\gamma f \sigma(h_{t+1})}{c_{t+1}^y} = \lambda_{1t+1} \quad (22)$$

By combining (17)-(19), I obtain the Euler equation given by (8). By combining (17) and (22), I can derive the following equation:

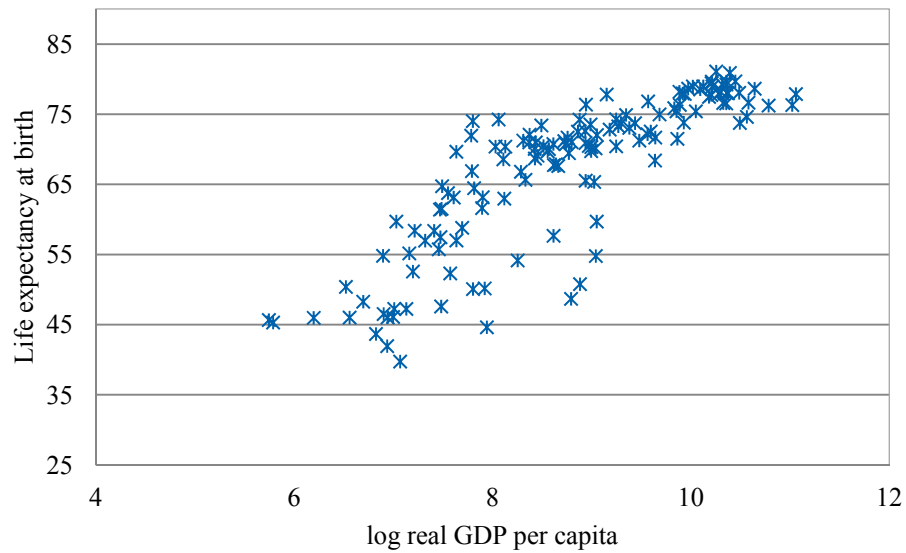
$$\frac{c_{t+1}^y}{\gamma f \sigma(h_{t+1}) c_t^y} = \frac{\lambda_{1t}}{\lambda_{1t+1}} \quad (23)$$

Equation (23) gives the *intergenerational marginal rate of substitution of consumption*: the marginal utility of consumption of the current young adult to the total marginal utility of young adults in the next period as valued by their parents must be equal to the ratio of the shadow prices associated with the budget constraints between the two periods. Now, I can substitute (17), (20) and (23) into (21) to arrive at the following equation:

$$\begin{aligned} & \frac{x_t^{1-\eta} h_t^{1-\eta}}{c_t^y \eta \xi} \\ = & \gamma \sigma'(h_{t+1}) U_{t+1} + \frac{\gamma \sigma(h_{t+1}) \beta w_{t+1} h_{t+1}^{\beta-1} m^{1-\beta}}{c_t^y} + \frac{\gamma f \sigma(h_{t+1})}{c_t^y} \frac{x_{t+1}}{h_{t+1}} \frac{1-\eta}{\eta} \end{aligned}$$

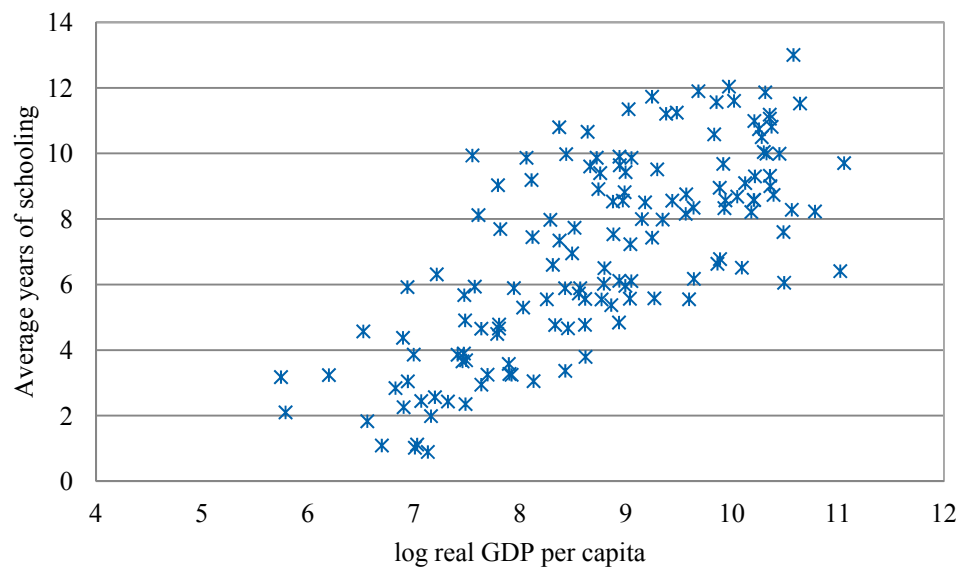
Equation (15) follows directly by imposing the steady-state conditions to the above equation.

Figure 1. Life expectancy at birth and log real GDP per capita



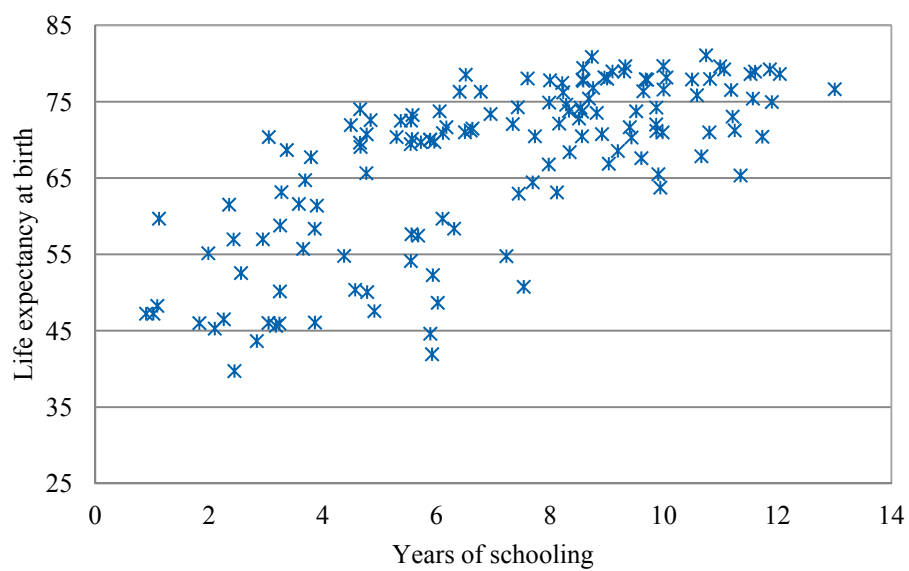
Source: the real GDP per capita, Penn World Table 6.3; life expectancy at birth in 2000, the World Bank.

Figure 2. Average years of schooling and log real GDP per capita



Source: the real GDP per capita, Penn World Table 6.3; the average years of schooling, Barro-Lee (2010).

Figure 3. Life expectancy at birth and average years of schooling



Source: Life expectancy at birth in 2000, the World Bank; the average years of schooling in 2000, Barro-Lee (2010).

Figure 4. Illustration of a shift in the survival probability distribution

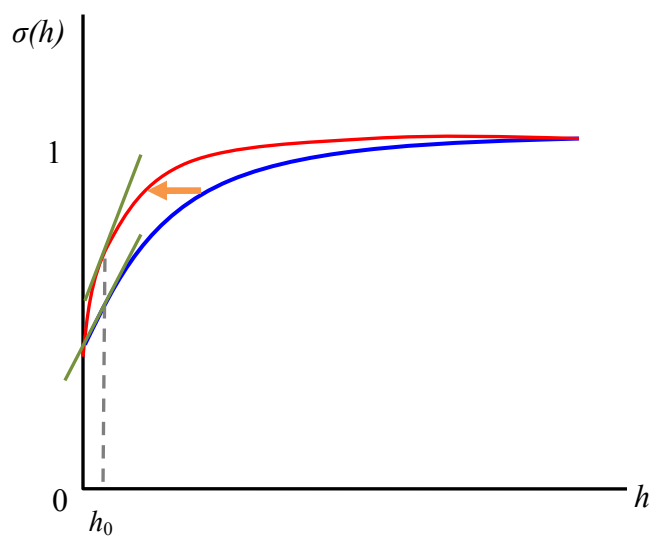


Figure 5. Steady state equilibrium and effects of changes in altruism and health technology

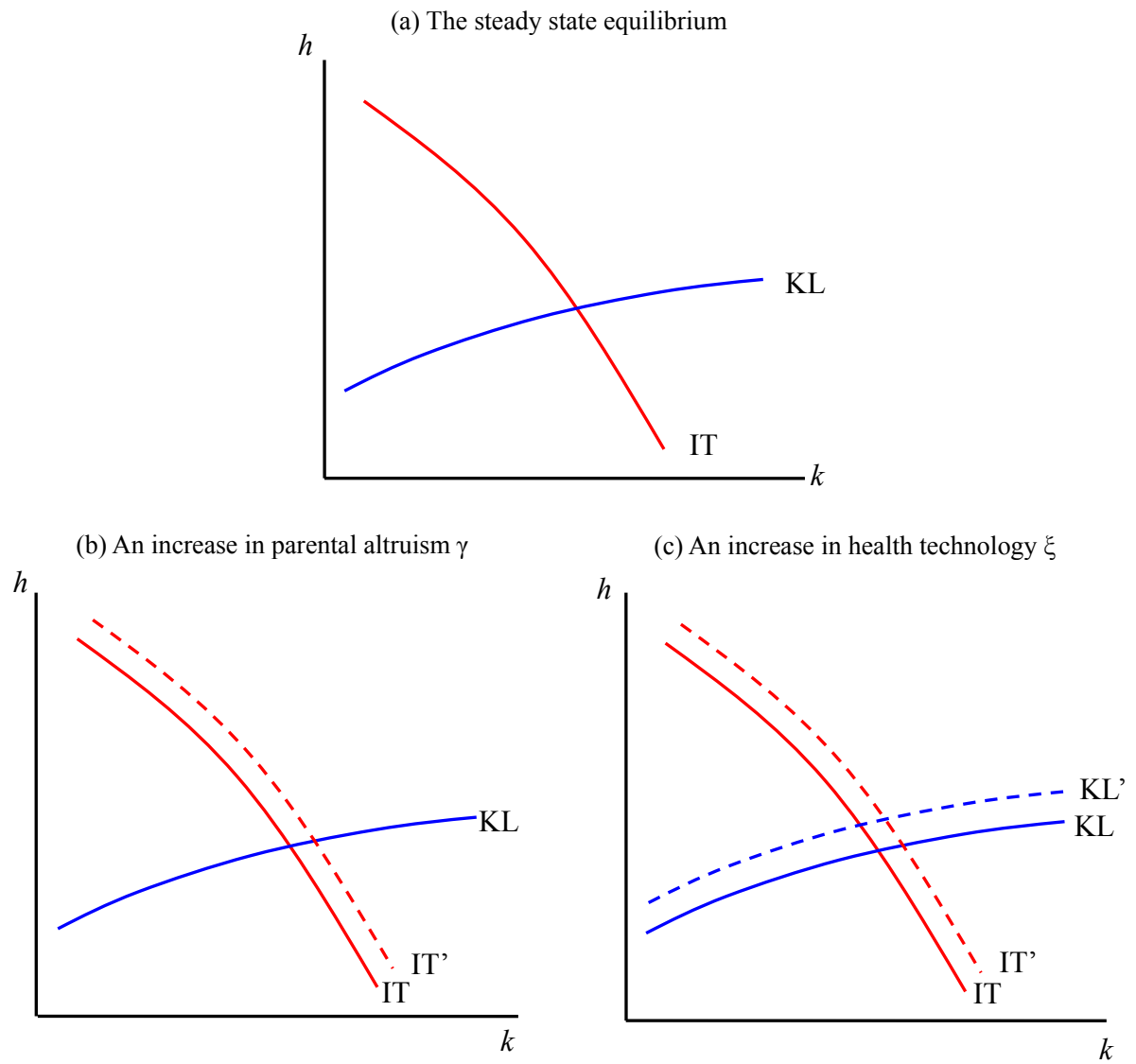


Table 1. Survival rates and life expectancies: by income level

Real GDP per capita, % of U.S.	Ave real GDP per capita	Relative real GDP per worker (US=1)	Ave survival rate to age 25	Ave life expectancy at birth	Ave economic growth rate
<5%	1270.99	0.0409	0.8011	51.4720	0.3822%
5%-10%	2697.67	0.0951	0.8899	61.9029	1.4206%
10%-20%	5772.14	0.1911	0.9344	68.3628	2.4287%
20-30%	9406.93	0.2972	0.9450	70.0610	1.3258%
30-40%	14212.61	0.4490	0.9701	71.8586	2.2583%
40-50%	19136.66	0.6363	0.9790	74.1100	3.2900%
50-60%	21265.60	0.5859	0.9828	77.0050	2.2696%
60-70%	26671.01	0.7457	0.9882	78.1867	2.1951%
70-80%	29880.13	0.7961	0.9870	77.7433	2.3951%
80-90%	32198.51	0.7880	0.9875	78.4960	2.0823%
90% up	45171.92	1.0757	0.9838	76.3186	2.1546%

Source: Survival rates, the UN life tables; life expectancy, the World Bank; real GDP per capita and per worker, the

Penn World Table 6.3

Table 2. Survival rates, model and the data

Decile	Survival rate		Improved sanitation facilities coverage rate	Health capital
	Model	Data (σ)	θ	h
90-100	1	0.9849	0.9966	6.1758
80-90	1	0.9856	0.9988	6.1036
70-80	0.9998	0.9819	0.9861	3.6086
60-70	0.9895	0.9581	0.8749	1.6408
50-60	0.9371	0.9165	0.6468	0.6176
40-50	0.9549	0.9472	0.8265	0.8210
30-40	0.8714	0.9025	0.4901	0.1118
20-30	0.8503	0.8972	0.4567	0.0942
10-20	0.8271	0.8234	0.2173	0.0236
0-10	0.8271	0.7721	0.2644	0.0154

Table 3. Parameters, assumptions and targets

Parameters		Value	Target/Assumption/Explanation
<i>Preferences</i>			
Discount factor	ρ	2.3864	Annual time discount rate 0.05
Altruistic factor	γ	0.2362	True altruistic factor 0.8
<i>Production technology</i>			
Total factor productivity	A	7.4275	Calibrated from the model
Physical capital share	α	0.36	Capital income share for the U.S.
Health capital share	β	0.0595	Calibrated from the model
Health investment share	η	0.5	Assumption
Proxy of health technology scaling factor	b	10.8470	Estimated from NLS
Health technology scaling factor	ζ	10.8470	Equal to $b\theta$
<i>Other moments and parameters</i>			
Physical capital per worker	k	2.4916	Calculated from PWT6.3, year 2000
Mincerian return	ϕ_s	0.1	Mincerian return for the U.S., Psacharopoulos and Patrinos (2004)
Knowledge capital per worker	m	3.63	Calculated from Mincerian return
Health expenditure to total output ratio	x/y	0.1251	Average health expenditure to total GDP ratio, 1995-2000
Investment to output ratio	I/Y	0.2	Investment to total GDP ratio
Annual interest rate	r_a	0.0391	Calibrated from the model
Capital depreciation rate	δ	2.0067	Match the investment-to-output ratio of 0.2
Fertility rate	f	1.0187	Inverse of the survival rate at age 25
Survival rate at age 25	σ	0.9817	Year 2000 survival rate at age 25 from WHO
Mean of the survival probability function	μ	-0.9241	Estimated from NLS
Variance of the survival probability function	ε^2	1	Assumption

Table 4. Data

	Survival rate	Average output per worker	Years of schooling	Physical capital per worker	Improved sanitation facilities coverage rate
Decile	σ	y/y_{US}	s_y	k	θ
90-100	0.9849	0.9687	9.1236	2.4690	0.9966
80-90	0.9856	0.7509	9.8009	2.1066	0.9988
70-80	0.9819	0.5704	8.7778	1.8271	0.9861
60-70	0.9581	0.3568	7.3896	0.7682	0.8749
50-60	0.9165	0.2428	6.5645	0.4504	0.6468
40-50	0.9472	0.1852	6.1352	0.4260	0.8265
30-40	0.9025	0.1212	5.4038	0.1883	0.4901
20-30	0.8972	0.0757	4.8353	0.1634	0.4567
10-20	0.8234	0.0508	3.4814	0.0548	0.2173
0-10	0.7721	0.0269	2.6094	0.0276	0.2644

Sources of data: Survival rate at age 25, WHO; years of schooling, Barro and Lee (2010). Relative output and physical capital per worker, author's calculation using PWT 6.3 with the normalization that the real GDP per worker for the US in 2000 is equal to one. The sample size is 117 countries.

Table 5. Relative TFPs

Decile	Survival rate		Average output per worker y/y_{US}	TFP/TFP _{US}				
	Data	Model		Model K	Model KM	Model KMX	Model KMH	Model KMH, $x'_{US}=x_{US}/2$
90-100	0.9849	1.0000	0.9687	0.9718	1.4360	1.4462	1.4466	1.4120
80-90	0.9856	1.0000	0.7509	0.7976	1.1566	1.1668	1.1669	1.1365
70-80	0.9819	0.9999	0.5704	0.6378	0.8479	0.8765	0.8774	0.8546
60-70	0.9581	0.9902	0.3568	0.5449	0.8292	0.8695	0.8784	0.8555
50-60	0.9165	0.9346	0.2428	0.4495	0.5692	0.6144	0.6351	0.6185
40-50	0.9472	0.9559	0.1852	0.3497	0.5361	0.5748	0.5832	0.5681
30-40	0.9025	0.8708	0.1212	0.3071	0.4805	0.5371	0.5671	0.5523
20-30	0.8972	0.8503	0.0757	0.2020	0.3324	0.3712	0.3940	0.3837
10-20	0.8234	0.8280	0.0508	0.2006	0.3500	0.3917	0.4399	0.4285
0-10	0.7721	0.8288	0.0269	0.1363	0.2563	0.2939	0.3252	0.3167
Relative TFP, top 10th and bottom 10th				7.1319	5.6018	4.9213	4.4486	4.4584

Table 6. Variance Decomposition

Variance	
$\log(y)$	1.3373
$\log(y_k)$	0.3129
$\log(y_{km})$	0.4659
$\log(y_{kmh})$	0.5743
Success (success=Var(log(y_X))/Var(log(y)))	
Model K	0.2340
Model KM	0.3484
Model KMH	0.4294

Table 7. Data versus model: by decile

Decile	Survival rate		Output per worker ($y_{US}=1$)		Physical capital per worker		Health expenditure per worker		Health capital per worker	
	data	model	data	model	data	model	data	model	estimate	model
90-100	0.9849	1.0000	0.9687	1.0028	2.4690	3.2261	0.0565	0.1407	6.5980	16.4438
80-90	0.9856	1.0000	0.7509	0.7386	2.1066	2.4625	0.0556	0.1036	6.5257	12.1648
70-80	0.9819	0.9999	0.5704	0.5269	1.8271	1.7624	0.0336	0.0739	3.8417	8.4569
60-70	0.9581	0.9902	0.3568	0.4111	0.7682	1.3225	0.0186	0.0571	1.6785	5.1452
50-60	0.9165	0.9346	0.2428	0.3048	0.4504	0.9535	0.0116	0.0478	0.5713	2.3541
40-50	0.9472	0.9559	0.1852	0.2075	0.4260	0.6816	0.0103	0.0302	0.8240	2.4267
30-40	0.9025	0.8708	0.1212	0.1689	0.1883	0.4965	0.0033	0.0270	0.0943	0.7622
20-30	0.8972	0.8503	0.0757	0.0926	0.1634	0.3322	0.0032	0.0096	0.0776	0.2355
10-20	0.8234	0.8280	0.0508	0.0872	0.0548	0.2804	0.0027	0.0101	0.0152	0.0560
0-10	0.7721	0.8288	0.0269	0.0496	0.0276	0.1717	0.0013	0.0047	0.0106	0.0389

Table 8. Countries with real GDP per worker of less than 10 percent of the U.S.: by region

Region	Average output per worker	Survival rate, data	Survival rate, model	Improved sanitation facilities coverage rate	Years of schooling	TFP/TFP _{US}			
	y/y_{US}					Model K	Model KM	Model KMX	Model KMH
East Asia and the Pacific – all	0.0629	0.9066	0.8378	0.3750	5.0849	0.2134	0.3478	0.3912	0.4215
East Asia and the Pacific. ₁ *	0.0585	0.8981	0.8364	0.3555	4.8264	0.2097	0.3537	0.3976	0.4302
Latin America and the Caribbean	0.0693	0.9043	0.8566	0.4369	5.1680	0.1539	0.2346	0.2603	0.2772
South Asia – all	0.0659	0.8772	0.8294	0.2979	3.1429	0.2176	0.4039	0.4515	0.4951
South Asia. ₁ **	0.0563	0.8810	0.8295	0.3253	2.9456	0.1975	0.3720	0.4181	0.4554
Sub-Saharan Africa	0.0387	0.7863	0.8293	0.2454	2.9118	0.1707	0.3087	0.3493	0.3888

* Excluding China; ** excluding India.